

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 12

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>Any 4 from:</i> • Sludge from primary sedimentation is digested by anaerobic microbes in a warm tank for about four weeks • Kills most pathogens • Reduces odour and volume of the sludge • Produces biogas (mainly methane with some CO₂) used as fuel to heat the digester / generate electricity • Digested sludge has a better C:N ratio and can be used as agricultural fertiliser	4
01.2	• Heavy metals from urban wastes / road runoff may be present and absorbed by crops - reduced by adding lime to lower metal solubility • Remaining pathogens - destroyed by heat treatment of the sludge before use • Timing: spreading in the growing season covers crops; in winter vehicles compact wet soils - apply at suitable times / when soils are dry • Odour / public objection - digestion reduces odour	4
01.3	• Foul waste is a small volume needing full treatment so smaller, cheaper treatment works can be built; surface water needs little treatment • The treatment works is not affected by weather-related fluctuations in flow (storms) so storm overflows of untreated sewage are avoided	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 270-273; AQA spec 3.4.3.2.9.	
02.1	• Spearman's rank correlation coefficient • Mann-Whitney U test • Chi-squared test • t-test	4
02.2	• Chi-squared compares frequencies / counts in categories • Dissolved oxygen is continuously variable measured data, so a t-test or Mann-Whitney U test is needed	2

Question	Marking guidance	Marks
02.3	• There is always some remaining uncertainty / a probability that the result was produced by chance • Statistics only give a level of confidence (eg 95%, 99%) that the result is significant	2
02.4	• Bars drawn above and below each mean showing plus and minus one standard deviation • If the bars of two means overlap a lot the difference is probably not significant; if they do not overlap it probably is	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 422-423; AQA spec 3.7.2.3.	
03.1	• Shale has very low permeability so the 'tight' oil / gas cannot flow through the rock to a well • Fluid is pumped down at high pressure to open fissures in the shale along which oil / gas can flow to the recovery well • Sand grains (to hold fissures open) and solvents are pumped in to increase the recovery rate	3
03.2	• Natural gas may enter aquifer water • Injected chemicals may reach aquifers or the surface; toxic metals in the rock may be mobilised • Large volumes of water needed • Earth tremors as fracking releases natural tensions in the crust (though not tremors that could not occur naturally) • Reduced by: collection, treatment and re-use of waste water; well casing standards; restricting sites away from aquifers / sensitive areas; monitoring	5
03.3	• Extraction is expensive and energy-intensive (quarrying and hot water treatment, steam / solvent / combustion in-situ; heating shale to release oil) so it is not economically viable at current prices • Reserves only include what is economically exploitable now; high environmental impacts (land take, water, CO₂) also limit exploitation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 184, 189-190; AQA spec 3.3.4.1.	
04.1	• pH 5.6 • Atmospheric carbon dioxide dissolves in rainwater to form (weak) carbonic acid	2
04.2	• Limestone bedrock (calcium carbonate) is alkaline / basic and neutralises the acids as water percolates through it, so the lake pH stays high (7.2) • Granite does not neutralise acids so lake water in A is acidic (4.6) • Low pH mobilises / dissolves aluminium ions from the soil and rock (0.42 mg dm⁻³ in A) which are toxic to fish (damage gills) • Acids also damage fish eggs / gills (exposed living cells) and dissolve the exoskeletons of invertebrate food; lichens and other food organisms lost	4
04.3	• Corrodes metals: railway lines, railings, pipes, pylons, overhead power lines • Dissolves / weakens limestone and marble buildings and statues (eg York Minster)	2
04.4	• Sulfur dioxide is oxidised (by ozone / other oxidants) to sulfur trioxide • Sulfur trioxide dissolves in water droplets to form sulfuric acid (SO₂ alone gives the weaker sulfurous acid)	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 250-252; AQA spec 3.4.3.2.2.	
05.1	• Its total annual discharge	1
05.2	• Public supplies cannot match consumer demand / the water is often cleaner than rivers • It also reduces urban flooding after heavy rain; surplus can be used for aquifer recharge	2
05.3	• The reservoir is created in the estuary so no farmland / settlements need to be flooded • But it changes important inter-tidal habitats, obstructs shipping and can be polluted by activities anywhere in the river's catchment	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-123; AQA spec 3.2.2.5.	
06.1	• Metal-rich lumps 5-10 cm across on the deep seabed (4000-5000 m), containing about 30% manganese plus nickel, copper, cobalt, iron, titanium • Exploitation would be very expensive (remotely operated collectors, pumping to the surface) / no viable technology yet • International agreement on ownership of the seabed and its resources is needed	3
06.2	• Collectors disturb the seabed and kill benthic organisms living on it • Separating nodules from sediment creates plumes that increase turbidity with unknown consequences • Re-deposition of sediment covers and kills filter feeders and other benthic organisms; slow recovery in the deep sea	3
06.3	• Used in small amounts in many essential technologies (smartphones, hybrid vehicles, wind turbines, fibre optics, LEDs) • Exploitable land deposits are limited and mainly in China; chemical separation is complex and produces large amounts of toxic waste • Bacterial adsorption: dissolved rare earth ions adsorb onto the cell surfaces of bacteria (<i>Bacillus subtilis</i>) and are washed off at specific pHs to separate them • Deep sea sediments contain relatively high concentrations but no viable extraction method yet	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 145-146; AQA spec 3.2.3.7.	
07.1	• The time taken for half of a pollutant (released into the environment) to break down / degrade	1
07.2	• DDT is liposoluble so it is stored in fat / can pass through phospholipid cell membranes and is not excreted (nitrates are water soluble and excreted easily) • DDT is persistent (long half-life) so it is not broken down before it is passed to the next trophic level • So DDT bioaccumulates in each organism and becomes more concentrated at each successive trophic level	3

Question	Marking guidance	Marks
07.3	<i>Any 2 from:</i> • The gas is denser than air so it settled close to the ground / did not rise and disperse • Low wind velocity (at night) meant it was not dispersed / diluted • Short half-life so it broke down before travelling far	2
07.4	• High solubility means nitrates are easily dispersed / leached / carried by water so they affect a larger area (eg rivers, aquifers, the sea) • But dispersal dilutes the pollutant so the concentration at any one point may be lower	2
07.5	<i>Any 2 from:</i> • Noise / sound • Heat / thermal pollution • Ionising radiation • Light	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240; AQA spec 3.4.1.	
08.1	• Landfill fell from 79% to 14% (large decrease) • Incineration rose from 9% to 42% and recycling / composting rose from 11% to 43% (now the largest)	2
08.2	<i>Any 3 from:</i> • Organic matter decays anaerobically releasing methane (and CO₂) - greenhouse gases • Toxic leachate may leak into groundwater / rivers from poorly managed sites • Uses large areas of land; loss of habitats / farmland; contamination prevents later use • Resource value of metals, glass, plastics is lost • Transport noise / congestion / litter / pests / odour	3
08.3	<i>Any 4 from:</i> • Polymer / clay liner to prevent escape of leachate • Collection and treatment of leachate • Collection of methane and use as a fuel • Impermeable cap (clay and polymer) when complete to stop water entering • Regular covering with soil (pests, odour); deodorising sprays • Perimeter fence to trap wind-blown litter • Separation and recording of waste types; reactive / flammable wastes not dumped together	4
08.4	• Mixed / contaminated / dirty wastes (plastics, dirty paper) are difficult and costly to separate and recycle but can be burnt with energy recovery	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 280-282; AQA spec 3.4.3.2.12.	
09.1	• Wheel vibration: track polishing / grinding, sound-absorbing ballast, sound-absorbing suspension • Pantograph: aerodynamic fairing • Wheel squeal: lubrication of wheels / track on corners • Braking: composite-material brakes instead of cast iron	4
09.2	• Roads pass through / near almost all residential areas and traffic is continuous through the day, unlike airports or railways which affect limited areas	1

Question	Marking guidance	Marks
09.3	<i>Any 3 from:</i> • Sound-absorbing road surfaces (porous asphalt / rubber-crumb asphalt) • Acoustic fences, embankments / baffle mounds, walls or dense tree planting • Traffic management for free flow (less acceleration noise); re-routing traffic away from residential areas • Double glazing / acoustic insulation of buildings • Vehicle design: better aerodynamics, engine insulation, quieter exhausts, quieter tyres	3
09.4	<i>Any 2 from:</i> • Air bubble curtains around the pile driver absorb noise • Acoustic survey to detect cetaceans first and delay work if present; increase volume gradually (soft start) so animals can leave • Drilling instead of pile-driving	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 292-294; AQA spec 3.4.3.2.13.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Stages of scientific method illustrated by ozone: identification (Antarctic UV data), hypothesis (Rowland-Molina 1974), data collection / analysis (Dobson units, satellites, balloons confirming CIO), proposed solutions, enactment (Vienna 1985, Montreal 1987) • Evidence of success: ozone hole area and depth stabilised and beginning to shrink; CFC concentrations in the atmosphere falling; UV levels not rising further; global ozone loss limited to about 4% • Reasons for success: international recognition of serious consequences (skin cancer); agreement of nearly every country; legally binding targets with a fund for developing countries; relatively few producers / products; alternatives (HCFCs, HFCs, pump sprays, hydrocarbons) already available and cheap; no lifestyle change needed • Limits: persistence of CFCs means recovery takes decades; some ODSs still used; HFC replacements are greenhouse gases • Climate change contrast: greenhouse gases come from almost every human activity (energy, transport, agriculture, industry) in every country; fossil fuels underpin economies so control requires major lifestyle and economic change • Agreements (Kyoto, Paris) have weaker / non-binding targets, disputes between developed and developing nations; emissions still rising; natural fluctuations, time delays and tipping points make cause-effect less clear to the public • Judgement: ozone control was a technical substitution problem; climate change is a systemic one, so the ozone success is only partly transferable	9
10.2	• Vienna Convention for the Protection of the Ozone Layer - no legally binding goals	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-113, 105-106; AQA spec 3.2.1.3, 3.2.1.2.	

Question	Marking guidance	Marks
11.1	Mineral resources are non-renewable. Discuss how an understanding of geological processes and the development of new technologies can extend the period for which society can rely on mineral resources. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Why non-renewable: formed too slowly; geological concentration processes - hydrothermal, metamorphic, Proterozoic sediments, alluvial, evaporites, secondary enrichment, biological sediments • Understanding processes guides exploration (batholith margins, river gravels, ancient seas) and remote sensing / geophysical techniques • Reserves, resource, stock; Lasky's principle; cut-off ore grade and its drivers • Extraction technologies: mechanisation, bioleaching, phytomining, polymer and bacterial adsorption, seabed nodules • Reducing demand: recycling and its difficulties, Cradle to Cradle, design for disassembly, linear vs circular material flows, dispersal of metals in use • Environmental limits: impacts of mining and their control; energy costs rising with lower grades; conflicts • Judgement: technology can extend supplies substantially but only within energy and environmental constraints; circular economy essential Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 130-147, 373-379; AQA spec 3.2.3, 3.6.3-3.6.4.	25

Question	Marking guidance	Marks
11.2	Discuss how knowledge of the properties of pesticides can be used to reduce their environmental impacts while maintaining food production. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Properties: toxicity (LD50, MDAF, MDNF), specificity, persistence, liposolubility/water solubility, mobility, mode of action (contact/systemic), synergism • Bioaccumulation and biomagnification explained by persistence + liposolubility; DDT and birds of prey as a case study • Comparison of organochlorines, organophosphates, pyrethroids, neonicotinoids with their relative advantages and disadvantages • Reducing impact: restrictions/bans, choice of non-persistent and specific pesticides, systemic pesticides and seed coatings, application timing (night, still days, not in flower), droplet size • Alternatives: cultural control (rotation, barrier/companion crops), biological control, predator habitats, sterile males, pheromone traps, genetic resistance, GM crops, integrated pest management • Trade-offs: yield, cost, human health (organophosphates), resistance evolution, pollinator services; risk:benefit; sustainability and the circular economy • Judgement: no pesticide is perfectly specific; combination of property-based selection and integrated control is most sustainable Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240, 262-266, 314-320; AQA spec 3.4.3.2.8, 3.5.1.1, 3.5.1.5.	25